

NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
Higher 1

CHEMISTRY

Paper 1 Multiple Choice

8872/01

27 September 2011

50 minutes

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, class and tutor's name on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

This document consists of **12** printed pages.

[Turn over

Section A

For each question there are four possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct.

- 1 Glucose, $C_6H_{12}O_6$, is a simple sugar and an important source of energy. An adult human can be considered to have a daily diet of 1.80kg of glucose.

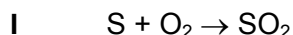
What mass of carbon dioxide does a person produce each day if all the glucose eaten is digested and oxidised?

- A 0.267 kg
- B 0.800 kg
- C 1.32 kg
- D 2.64 kg

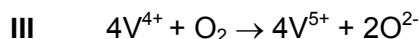
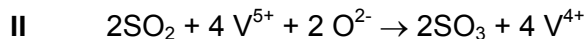
- 2 The Contact Process is the current method of producing sulfuric acid in the high concentrations needed for industrial processes.

The process can be divided into four stages:

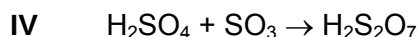
Making the sulfur dioxide:



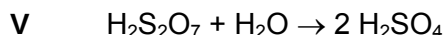
Conversion of sulfur dioxide to sulfur trioxide using V_2O_5 :



Sulfur trioxide formed is added to sulfuric acid which gives rise to oleum (disulfuric acid):



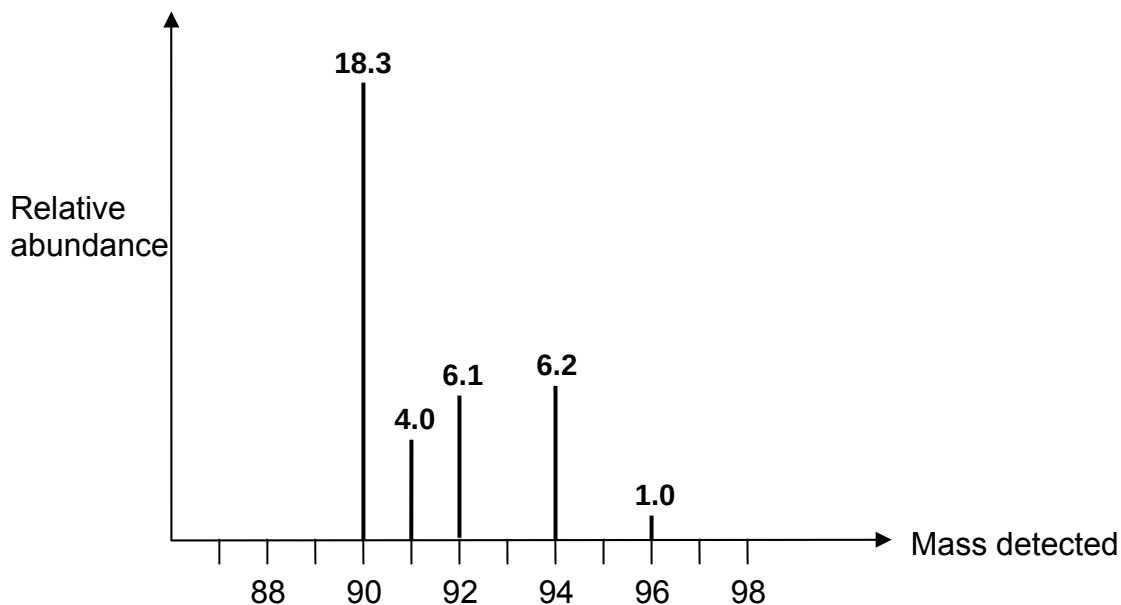
The oleum then is added to water to form concentrated sulfuric acid:



Which statement is **not** correct?

- A In reaction II, only sulfur is being oxidised.
- B In reaction IV, H_2SO_4 is an oxidising agent.
- C Sulfur is being oxidised from 0 to + 6 through the entire process.
- D The regeneration of V^{5+} suggests that it is possibly a catalyst for part of the process.

- 3 Zirconium (Zr) is a strong transition metal used in refractory. The principle source of Zirconium is the mineral Zircon. A sample of Zircon was analyzed with a mass spectrometer and the following is a portion of the mass spectrum obtained.

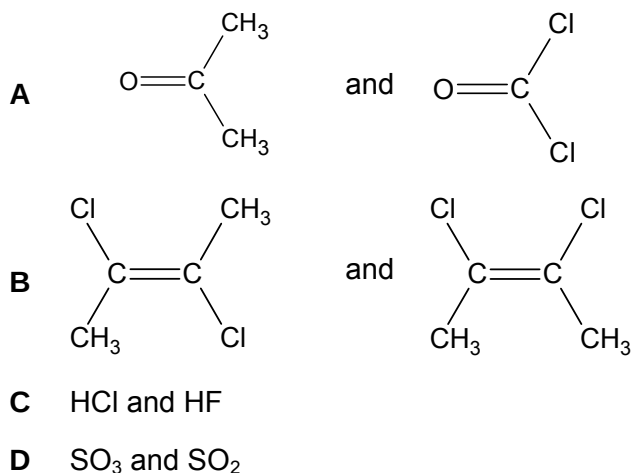


What is the relative atomic mass of Zr in the sample?

- A 91.1
 B 91.3
 C 93.0
 D 32.5
- 4 Which of the following ions contains more electrons than protons and more neutrons than protons?

- A $^{10}\text{B}^-$
 B $^{17}\text{O}^+$
 C $^{17}\text{F}^-$
 D $^{35}\text{Cl}^{2-}$

- 5 In which one of the following pairs does the first molecule have a larger dipole than the second?



- 6 Ammonia is a weak base that can react with aqueous hydrochloric acid. Which of the following bonds is **not** present in the product when molar quantities of the two reactants are mixed together?

- A** Hydrogen bond
- B** Coordinate bond
- C** Covalent bond
- D** Ionic bond

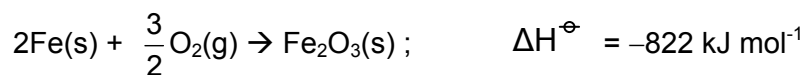
- 7 Which of the following compounds contain giant lattice structure?

- A** BaO, AlF₃, SiO₂
- B** K₂O, BCl₃, Cr₂O₃
- C** MgO, Fe₂O₃, P₄O₁₀
- D** Na₂O, SiO₂, SO₃

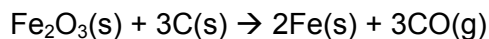
- 8 Which of the following describes a phenomenon which **cannot** be explained by hydrogen bonding?

- A** Ice has a lower density than water at 0°C.
- B** Nitrobenzoic acid is more volatile than 4-nitrobenzoic acid.
- C** Ethanoic acid molecules form dimers when dissolved in benzene.
- D** The boiling point of alcohol increases with increasing relative molecular mass.

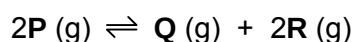
- 9 The standard enthalpy change for two reactions are given by the equations:



What is the standard enthalpy change for the following reaction?



- A + 712 kJ mol⁻¹
 B + 492 kJ mol⁻¹
 C – 492 kJ mol⁻¹
 D – 712 kJ mol⁻¹
- 10 For which of the following is the lattice energy likely to have the greatest numerical value?
- A lithium fluoride
 B lithium iodide
 C rubidium chloride
 D sodium chloride
- 11 Compound **P** decomposes on heating according to the following equation:

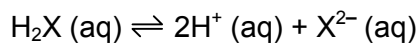


When 5 moles of **P** were put into a 1 dm³ vessel and heated, the equilibrium mixture contained 0.6 moles of **Q**.

What is the numerical value of the equilibrium constant, K_c ?

- A $\frac{(0.6)^2 \times 0.3}{5^2}$
 B $\frac{(0.6)^2 \times 1.2}{5^2}$
 C $\frac{0.6 \times (0.3)^2}{3.8^2}$
 D $\frac{0.6 \times (1.2)^2}{3.8^2}$

- 12 The equilibrium equation for the dissociation of a weak acid, H_2X , may be written as below:



What is expression for the acid dissociation constant, K_a ?

- A $\frac{[2\text{H}^+][\text{X}^{2-}]}{[\text{H}_2\text{X}]}$
- B $\frac{[\text{H}^+][\text{X}^{2-}]}{[\text{H}_2\text{X}]}$
- C $\frac{[\text{H}^+]^2[\text{X}^{2-}]}{[\text{H}_2\text{X}]}$
- D $\frac{[2\text{H}^+]^2[\text{X}^{2-}]}{[\text{H}_2\text{X}]}$

- 13 The value of the ionic product of water, K_w , varies with temperature as shown.

Temperature / °C	$K_w / \text{mol}^2 \text{dm}^{-6}$
25	1.0×10^{-14}
62	1.0×10^{-13}

Which of the following statements is true?

- A The ionic dissociation of water is exothermic.
- B The pH of water is lower at 62°C than at 25°C.
- C Water is not a neutral liquid at 62°C.
- D None of the above

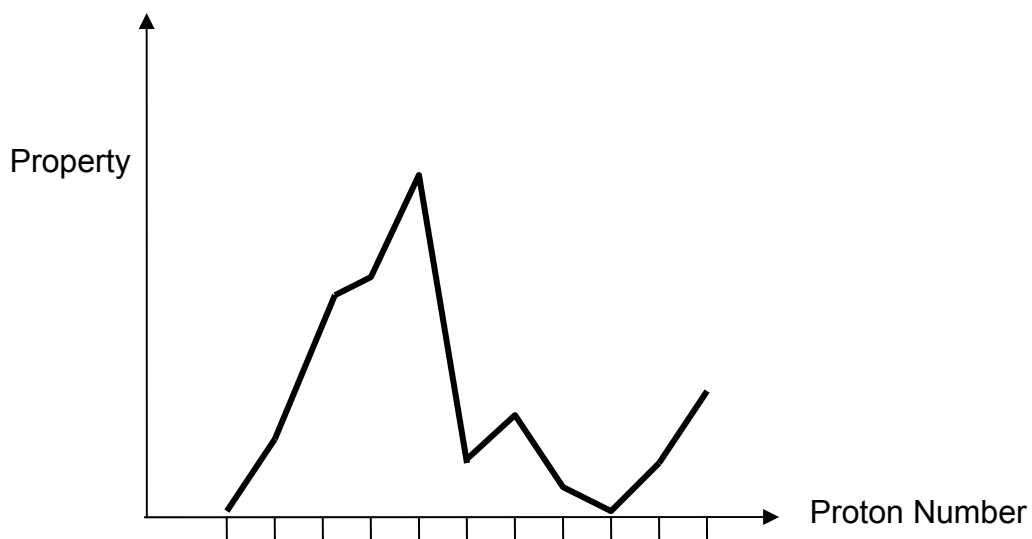
- 14** Carbon dating is a radiometric dating method used in archaeology. Living organisms contain the naturally occurring carbon-14 isotope at the atmospheric level due to the intake of carbon dioxide during photosynthesis. When an organism dies, the level of the carbon-14 isotope slowly declines from the atmospheric level due to radioactive decay. Hence by comparing the level of carbon-14 remaining in organic remains against the atmospheric level, one can deduce the estimated age of the sample.

Given that the half life of carbon-14 is 5568 years, deduce the age of an animal carcass that was found to contain a level of carbon-14 that is 10% of the atmospheric level.

- A** 550 years
B 16 700 years
C 17 800 years
D 22 300 years
- 15** In which one of the following do the species follow an increasing trend in size?
- A** B, C, N
B Cl^- , Ar, K^+
C Mg^{2+} , Na^+ , O^{2-}
D I^+ , Br^+ , Cl^+
- 16** Separate samples of Mg, MgCl_2 and MgO are respectively added to 3 beakers **I**, **II** and **III** containing deionized water. Which of the following gives the correct colour of the solution when a few drops of universal indicator is added to each of the beakers **I**, **II** and **III**?

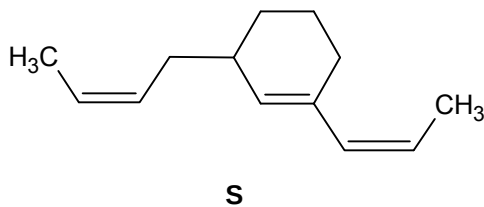
	<u>Beaker I</u>	<u>Beaker II</u>	<u>Beaker III</u>
A	Red	Orange	Purple
B	Red	Yellow	Blue
C	Purple	Orange	Blue
D	Purple	Yellow	Purple

- 17 The following graph shows the variation of a property of some elements, with proton number 10 to 20, with proton number.



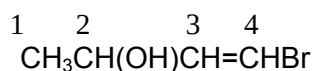
What is the property?

- A First ionisation energy
 - B Electrical conductivity
 - C Melting point
 - D Maximum oxidation state
- 18 What is the number of geometric isomers that compound **S** can have?



- A 2
- B 4
- C 6
- D 8

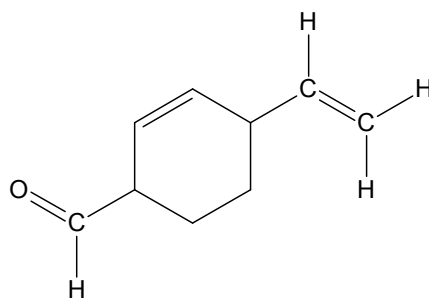
19 Compound T has the following structure



Which statement is **not** correct for compound T?

- A The terminal carbon atom (4) has a trigonal planar geometry.
- B T reacts with acidified potassium dichromate(VI).
- C T reacts with 2,4-dinitrophenylhydrazine.
- D T exhibits geometrical isomerism.

20 Compound V was reacted with hydrogen gas in the presence of palladium catalyst. What is the total number of sigma and pi bonds found in the product formed?



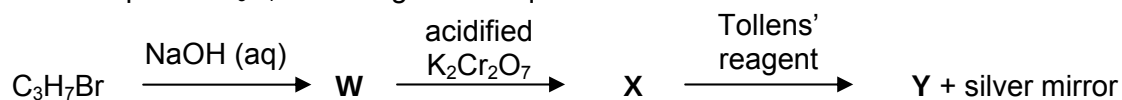
Compound V

	No of sigma bonds	No of pi bonds
A	21	3
B	25	2
C	26	1
D	28	0

21 Which property of benzene may be directly attributed to the stability associated with its delocalised electrons?

- A It has a low boiling point.
- B It does not conduct electricity.
- C Its enthalpy change of formation is positive.
- D It tends to undergo substitution rather than addition reactions.

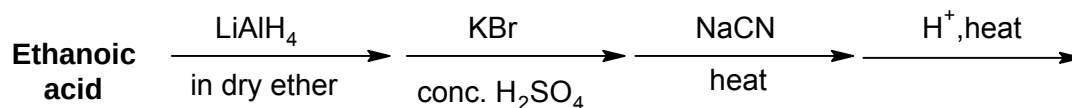
- 22 The compound C_3H_7Br undergoes a sequence of reactions as follows.



What could be the formulae for **W**, **X** and **Y**?

- | | W | X | Y |
|----------|------------------|-----------------|------------------|
| A | $CH_3CH_2CH_2OH$ | CH_3CH_2CHO | $CH_3CO_2^-$ |
| B | $CH_3CH_2CH_2OH$ | CH_3CH_2CHO | $CH_3CH_2CO_2^-$ |
| C | $CH_3CH_2CH_2OH$ | $CH_3CH_2CO_2H$ | CH_3CH_2CHO |
| D | $CH_3CH(OH)CH_3$ | CH_3COCH_3 | $CH_3CO_2^-$ |
- 23 Which compound reacts with its own oxidation product to give a sweet-smelling product?
- A** propanal
B propanone
C propan-1-ol
D propan-2-ol
- 24 In the addition reaction of carbonyl, CN^- ions from HCN are added across the carbonyl centre.
- In a similar reaction, $CH_3CH_2^-$ ions are generated from special reagents called Grignard reagents, CH_3CH_2MgBr .
- Which compound can be made from an aldehyde and CH_3CH_2MgBr ?
- A** $(CH_3)_2C(OH)CH_2CH_3$
B $CH_3CH(OH)CH_2CH_2CH_3$
C $(CH_3CH_2)_3C(OH)$
D $(CH_3CH_2)_2CH(OH)$

- 25 Ethanoic acid is reacted in the following sequence of reactions. What is the final product?



- A** CH_3CH_2COOH
B $CH_3CH(OH)COOH$
C $CH_3CH(Br)COOH$
D $CH_3CH_2CH_2NH_2$

Section B

For each of the questions in this section one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements which you consider to be correct).

The responses **A** to **D** should be selected on the basis of

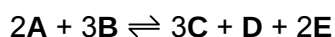
A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

26 Which particle would, on losing an electron have a half-filled set of orbitals?

- 1** N^-
- 2** Cr
- 3** O^+

27 In the following reaction



3 mol of **A** and 4 mol of **B** were initially present in a reaction vessel. It was found that 1.5 mol of **C** was present at equilibrium. Which of the following is also present at equilibrium?

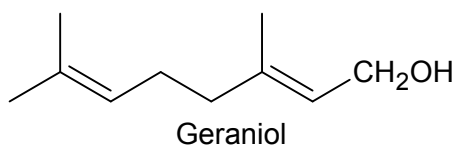
- 1** 2.0 mol of **A**
- 2** 2.5 mol of **B**
- 3** 1.0 mol of **E**

The responses **A** to **D** should be selected on the basis of

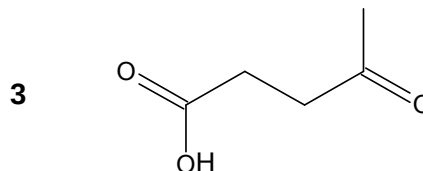
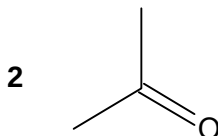
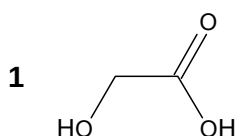
A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

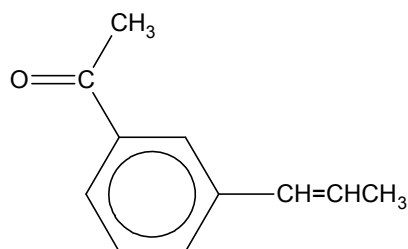
28 Geraniol has a rose-like odour and is commonly used in perfumes.



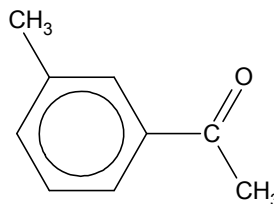
Which of the following is formed when geraniol reacts with hot acidified potassium manganate(VII)?



29 Which of the following reagents will give a positive test for **both** compound **F** and **G**?



Compound **F**



Compound **G**

- 1** Aqueous bromine
- 2** Hot acidified KMnO_4
- 3** 2,4-dinitrophenylhydrazine

30 $\text{CH}_2=\text{CHCH}_2\text{CH}(\text{OH})\text{CO}_2\text{H}$ is reacted with an excess of the following reagents. In which of the reactions is 1 mole of gas produced?

- 1** Hot acidified KMnO_4 ,
- 2** Na
- 3** Na_2CO_3

- 1 D
- 2 B
- 3 B
- 4 D
- 5 A
- 6 A
- 7 A
- 8 D
- 9 B
- 10 A
- 11 D
- 12 C
- 13 B
- 14 C
- 15 C
- 16 D
- 17 C
- 18 B
- 19 C
- 20 D
- 21 D
- 22 B
- 23 C
- 24 D
- 25 A
- 26 B
- 27 A
- 28 C
- 29 C
- 30 B

